

Qualification Pack



Power Plant Millwright Fitter

QP Code: PSS/Q0301

Version: 1.0

NSQF Level: 4

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri
New Delhi-110021 || email:akbhagatin@yahoo.co.in



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PSS/Q0301: Power Plant Millwright Fitter

Brief Job Description

The incumbent works on power plant machinery and mechanical equipment and components. This equipment may include turbines and internal combustion engines, power transmission assemblies, basic pneumatic systems, basic hydraulic systems, pumps, compressors, fans, fuel handling system, lubrication, cooling and exhaust systems, etc. Some components worked on include bearings, valves, drives.

Personal Attributes

Physically and mentally able to safely perform essential functions of the job. The candidate should be able to climb ladders, scaffolds, poles and towers of various heights. Also able to crawl and work in confined spaces such as attics, manholes and crawlspaces. The Candidate should be able to work for long working hours/nights, good eye visibility and ability to communicate to customer and resolve their problems. The individual should be ethical and well behaved.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N0301: Install power plant mechanical equipment at site](#)
2. [PSS/N0302: Perform maintenance activities on power plant equipment and machinery](#)
3. [PSS/N1336: Working effectively with others](#)
4. [PSS/N2001: Use basic health and safety practices for power related work](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Generation
Occupation	Plant & Equipment Maintenance
Country	India
NSQF Level	4
Credits	NA



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7233.38
Minimum Educational Qualification & Experience	10th Class with 1-2 Years of experience For ITI - 1 year on the job experience as Power Plant Mechanical Junior Fitter
Minimum Level of Education for Training in School	Same as Min.Ed for Job
Pre-Requisite License or Training	Certificate or Customised training on Mechanical Equipment or Machines Maintenance/Installation/Commissioning, MMA Welding and Oxy-Fuel Gas Cutting (Suggested but not Mandatory)
Minimum Job Entry Age	18 Years
Last Reviewed On	13/09/2017
Next Review Date	31/03/2022
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03487
NQR Version	1.0



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PSS/N0301: Install power plant mechanical equipment at site

Description

This unit covers the skills and knowledge required for installing a range of mechanical equipment at a power plant set-up including equipment such as machine tools, process control equipment, rotating mechanical equipment, conveyors, lifting and handling equipment hydraulic press, furnaces, auto / manual welding machines, shot blasting machines and processing plant machinery that have mechanical systems connected to them, in accordance with approved procedures.

Scope

This unit/task covers the following:

- Working safely
- Carry out a site check, prior to the installation
- Carry out a check on receiving the product for installation
- Prepare the product for installation
- Install the mechanical equipment

Elements and Performance Criteria

Working Safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (ppe) and other relevant safety regulations while performing fabrication and fitting operations
- PC3.** ensure work area is clean and safe from hazards
- PC4.** work safely in confined space, in trenches or excavated area,
- PC5.** handle large and heavy objects/loads and machines in a safe manner for self and others
- PC6.** work safely on high pressure line /system (steam, compressed air, hydraulic, etc.)
- PC7.** work safely on energized systems, steam and compressed air systems, etc
- PC8.** follow safety signages where displayed including road safety
- PC9.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC10.** obtain clearance to carry out the installation activities
- PC11.** provide safe access and working arrangements for the installation area
- PC12.** ensure safe isolation of services during the installation
- PC13.** dispose of waste items in a safe and environmentally acceptable manner
- PC14.** leave the work area in a safe condition and free from foreign object debris

Carry Out A Site Check, Prior To The Installation

To be competent, the user/individual on the job must be able to:

- PC15.** plan the installation activities in an efficient and appropriate manner

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- PC16.** survey and inspect the site and foundation for the following: Inspect the following: ensure that the site is accessible; ensure that site is free from obstructions or hazards; ensure the site is suitably prepared for the mechanical equipment installation to take place
- PC17.** ensure that appropriate utilities are available (e.g. gas, water, air, electricity)
- PC18.** ensure that required installation consumables are available
- PC19.** ensure that safety and environmental conditions can be met
- PC20.** obtain necessary permits to carry out the required work
- PC21.** check the installation job specification document-ation are available and correct Job specification documents: e.g. assembly drawings; layout drawings; contractual specifications; manufactures guidelines for installation; spares check and hand-over; manuals check and handover, etc.

Check and prepare the product for installation

To be competent, the user/individual on the job must be able to:

- PC22.** marking out of positioning and layouts
- PC23.** check and record for any physical damages to the machine/ equipment
- PC24.** check position and condition of anchor bolts/ foundation bolts
- PC25.** movement and positioning of equip-ment, using cranes or forklifts as per the layout
- PC26.** remove moisture absorbent bags, rust preventive, locking devices
- PC27.** fill oils for lubrication, hydraulic and other special oils
- PC28.** ensure the machine is clean

Install The Mechanical Equipment

To be competent, the user/individual on the job must be able to:

- PC29.** 'install the machine in accordance with manu-facturers" and site specifications'
- PC30.** instruction/guidance of the manufacturer/ customer and received through team supervisor/engineer
- PC31.** use the various installation tools and equipment as required. Instruments: straight edges and feeler gauges; spirit levels with appropriate accuracy; mandrels; dial test indicators; measuring instruments (meter tape, vernier caliper, micrometers, depth gauges); plumb lines and taut wires; customized gauges; right angle/square block
- PC32.** apply installation techniques like leveling, aligning, coupling and connecting in accordance with speci-fications
- PC33.** fill coolants, oil and other fluids as per specifications
- PC34.** ensure the site is cleaned and clear of all debris and left in safe state
- PC35.** all reports and documentation are completed correctly to required specifications
- PC36.** produce installations which comply with the equipment manufacturers operation specification /range
- PC37.** deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved
- PC38.** witness no-load trial run of the equipment and carry out check for proper functioning without load. Checks: system turns on; input and output voltage levels are being arrived at; hydraulics are working; pressure is building as per requirement; working offans, motors, a/cs, etc. and functioning properly; various sub-parts of the machinery functions; check oils and coolant;



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PC39. make adjustments, appropriate to the equipment being installed and seek guidance of the supervisor/ engineer when required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** importance of working in clean and safe environment
- KU4.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU5.** reporting structure, inter-dependent functions, lines and procedures in the
- KU6.** work area
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** documentation and related procedures applicable in the context of employment and work
- KU10.** importance and purpose of documentation in context of employment and work
- KU11.** procedures to be carried out before starting work on the installation
- KU12.** specific safe working practices, installation procedures and environmental regulations that must be observed
- KU13.** hazards associated with carrying out the installation of machinery and plant equipment and how can they be minimized
- KU14.** personal protective equipment to be used during the fabrication and fitting activities and where can it be obtained
- KU15.** types and sources of appropriate job specificationsJob specification documents: e.g. assembly drawings; layout drawings; contractual specifications; manufactures guidelines for installation; spares check and handover; manuals check and handover
- KU16.** common terminology used in installation of machinery and plant equipment
- KU17.** interpretation of drawings, standards, quality control procedures and specifications used for the installation including testing procedures
- KU18.** equipment to be installed, its operating procedures and function
- KU19.** methods of marking out the site for positioning of the equipment, and the tools and equipment used for this
- KU20.** methods of drilling holes for rag and expanding bolts (including the use of grouting and adhesives)
- KU21.** various mechanical fasteners that will be used, and their method of installation (e.g. threaded fasteners, special securing devices, masonry fixing devices)
- KU22.** torque loading requirements of the fasteners, and what to do if these loadings are exceeded or not achieved
- KU23.** correct tools, equipment, and fasteners for the installation activities



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- KU24.** types of tools and instruments used to position, secure and align the equipment (e.g. spanners, wrenches, crow bars, torque wrenches, engineers levels, alignment telescopes and laser devices) Instruments: straight edges and feeler gauges; spirit levels with appropriate accuracy; mandrels; dial test indicators; measuring instruments (meter tape, vernier caliper, micrometers, depth gauges); plumb lines and taut wires; customized gauges; multimeters; right angle/square block
- KU25.** techniques used to position, align, level and adjust the equipment
- KU26.** methods of lifting, handling and supporting the equipment during the installation activities
- KU27.** methods of connecting to mechanical power transmission devices (eg. belt and chain drives, couplings, clutches and brakes)
- KU28.** methods of connecting equipment to service supplies (eg. electrical, fluid power, compressed air oil and fuel supplies)
- KU29.** procedure for the safe disposal of waste materials
- KU30.** how to conduct any necessary checks to ensure the equipment integrity, functionality, accuracy, and quality of the installation Checks: setting working clearance; tensioning; checking level and alignment; making visual checks for completeness and freedom from damage; making sensory checks (sight, sound, smell, touch); ensuring that moving parts are guarded and clear of obstruction; checking torque settings of fasteners fitted at the site; ensuring locking devices are fitted to fasteners (where appropriate); ensure fulfillment of specific instruction in manufactures guidelines
- KU31.** how to recognize installation defects and how to address them appropriately Defects: leaks, poor seals, misalignment, ineffective fasteners, foreign object damage, contamination, vibration, etc.
- KU32.** importance of ensuring that the completed installation is free from dirt, and foreign object damage, and of ensuring that any exposed components or pipe ends are correctly covered/protected
- KU33.** care and control procedures for tools and equipment
- KU34.** problems that can occur with the installation operations, and how these can be overcome
- KU35.** fault-finding techniques to be used when the equipment fails to operate correctly
- KU36.** extent of own responsibility, and whom to report to in case there is a problems that is not getting resolved
- KU37.** various job related engineering drawings
- KU38.** knowledge of the mechanical equipment function and product
- KU39.** knowledge of component machining processes
- KU40.** dos and donts of operating and maintaining the machine

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in local language
- GS2.** convey and share technical information clearly using appropriate language
- GS3.** check and clarify task-related information
- GS4.** liaise with appropriate authorities using correct protocol



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- GS5.** communicate with people in respectful form and manner in line with organizational protocol
- GS6.** listen to questions and concerns of the customer and provide resolution in a respectful manner as per organizational guidelines
- GS7.** be well dressed and groomed
- GS8.** put forward one's point of view in a convincing manner
- GS9.** undertake numerical operations, geometry and calculations/ formulae Arithmetic: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
- GS10.** use appropriate measuring techniques
- GS11.** express numerical solutions to a degree of accuracy that is appropriate to the value being calculated Degree of accuracy: correct to three significant figures, correct to two decimal places, express a decimal fraction in standard form, express tolerance in terms of limits of size
- GS12.** use a calculator to raise a number to a power and determine square roots
- GS13.** plot and interpret straight line graphs
- GS14.** apply Pythagoras' theorem to perform calculations
- GS15.** define work, power and energy
- GS16.** define friction Friction: definition, explain coefficient of friction, explain how friction can be reduced, select materials that will rotate, or slide together with low frictional value
- GS17.** describe the relationship between temperature changes and changes in length
- GS18.** measure heights and angles at a site
- GS19.** participate in on-the-job and other learning, training and development interventions and assessments
- GS20.** clarify task related information with appropriate personnel or technical adviser
- GS21.** seek to improve and modify own work practices
- GS22.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS23.** perform basic operations in a computer like switching it on/off, using the mouse and keyboard, accessing files, opening, closing, creating and deleting folders, etc.
- GS24.** use email to communicate within the organization as per organization guidelines
- GS25.** retrieve and enter data using standard system forms and templates
- GS26.** take printouts of documents
- GS27.** identify problems with work planning, procedures, output and behavior and their implications
- GS28.** prioritize and plan for problem solving
- GS29.** communicate problems appropriately to others
- GS30.** identify sources of information and support for problem solving
- GS31.** seek assistance and support from other sources to solve problems
- GS32.** identify effective resolution techniques
- GS33.** select and apply resolution techniques
- GS34.** seek evidence for problem resolution
- GS35.** plan, prioritize and sequence work operations as per job requirements
- GS36.** organize and analyze information relevant to work



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- GS37.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS38.** undertake and express new ideas and initiatives to others
- GS39.** give inputs to modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS40.** participate in improvement procedures including process, quality
- GS41.** ones competencies in new and different situations and contexts to achieve more
- GS42.** exercise restraint while expressing dissent and during conflict situations
- GS43.** avoid and manage distractions to be disciplined at work
- GS44.** manage own time for achieving better results
- GS45.** work in a team in order to achieve better results
- GS46.** identify and clarify work roles within a team
- GS47.** communicate and cooperate with others in the team for better results
- GS48.** seek assistance from fellow team members
- GS49.** follow correct communication protocols with customers
- GS50.** work towards ensuring customer satisfaction and delight
- GS51.** contribute to customer satisfaction
- GS52.** meet customer needs for information and assistance
- GS53.** recognize and communicate limits of ones authority and ability in responding to customer expectations
- GS54.** collect and pass on accurate and timely customer feedback to appropriate company authorities
- GS55.** handle customer disgruntlement and dissatisfaction
- GS56.** apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working Safely</i>	8	31	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	1	3	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (ppe) and other relevant safety regulations while performing fabrication and fitting operations	1	3	-	-
PC3. ensure work area is clean and safe from hazards	-	2	-	-
PC4. work safely in confined space, in trenches or excavated area,	-	2	-	-
PC5. handle large and heavy objects/loads and machines in a safe manner for self and others	1	3	-	-
PC6. work safely on high pressure line /system (steam, compressed air, hydraulic,etc.)	1	2	-	-
PC7. work safely on energized systems, steam and compressed air systems, etc	1	2	-	-
PC8. follow safety signages where displayed including road safety	1	2	-	-
PC9. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	-	2	-	-
PC10. obtain clearance to carry out the installation activities	-	2	-	-
PC11. provide safe access and working arrangements for the installation area	-	2	-	-
PC12. ensure safe isolation of services during the installation	1	2	-	-
PC13. dispose of waste items in a safe and environmentally acceptable manner	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. leave the work area in a safe condition and free from foreign object debris	-	2	-	-
<i>Carry Out A SiteCheck, Prior To The Installation</i>	3	14	-	-
PC15. plan the installation activities in an efficient and appropriate manner	1	2	-	-
PC16. survey and inspect the site and foundation for the following: Inspect the following: ensure that the site is accessible; ensure that site is free from obstructions or hazards; ensure the site is suitably prepared for the mechanical equipment installation to take place	1	2	-	-
PC17. ensure that appropriate utilities are available (e.g. gas, water, air, electricity)	-	2	-	-
PC18. ensure that required installation consumables are available	-	2	-	-
PC19. ensure that safety and environmental conditions can be met	-	2	-	-
PC20. obtain necessary permits to carry out the required work	-	2	-	-
PC21. check the installation job specification document-ation are available and correct Job specification documents: e.g. assembly drawings; layout drawings; contractual specifications; manufactures guidelines for installation; spares check and hand-over; manuals check and handover, etc.	1	2	-	-
<i>Check and prepare the product for installation</i>	5	12	-	-
PC22. marking out of positioning and layouts	1	2	-	-
PC23. check and record for any physical damages to the machine/ equipment	1	2	-	-
PC24. check position and condition of anchor bolts/ foundation bolts	1	2	-	-
PC25. movement and positioning of equip-ment, using cranes or forklifts as per the layout	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. remove moisture absorbent bags, rust preventive, locking devices	1	1	-	-
PC27. fill oils for lubrication, hydraulic and other special oils	-	1	-	-
PC28. ensure the machine is clean	-	1	-	-
<i>Install The Mechanical Equipment</i>	8	19	-	-
PC29. 'install the machine in accordance with manufacturers' and site specifications'	1	3	-	-
PC30. instruction/guidance of the manufacturer/customer and received through team supervisor/engineer	1	2	-	-
PC31. use the various installation tools and equipment as required. Instruments: straight edges and feeler gauges; spirit levels with appropriate accuracy; mandrels; dial test indicators; measuring instruments (meter tape, vernier caliper, micrometers, depth gauges); plumb lines and taut wires; customized gauges; right angle/square block	1	1	-	-
PC32. apply installation techniques like leveling, aligning, coupling and connecting in accordance with specifications	1	2	-	-
PC33. fill coolants, oil and other fluids as per specifications	1	1	-	-
PC34. ensure the site is cleaned and clear of all debris and left in safe state	-	1	-	-
PC35. all reports and documentation are completed correctly to required specifications	1	1	-	-
PC36. produce installations which comply with the equipment manufacturers operation specification /range	1	2	-	-
PC37. deal promptly and effectively with problems within control, and seek help and guidance from the relevant people for problems that cannot be resolved	-	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. witness no-load trial run of the equipment and carry out check for proper functioning without load. Checks: system turns on; input and output voltage levels are being arrived at; hydraulics are working; pressure is building as per requirement; working offans, motors, acs, etc. and functioning properly; various sub-parts of the machinery functions; check oils and coolant;	1	2	-	-
PC39. make adjustments, appropriate to the equipment being installed and seek guidance of the supervisor/ engineer when required	-	2	-	-
NOS Total	24	76	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0301
NOS Name	Install power plant mechanical equipment at site
Sector	Power
Sub-Sector	Generation
Occupation	Thermal Power Plant Equipment Erection & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA



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PSS/N0302: Perform maintenance activities on power plant equipment and machinery

Description

The candidate will be required to maintain a range of mechanical equipment which could include gearboxes, machine tools, lifting and handling equipment, processing plant, production plant, engines, pumps, process control valves, compressors, transfer equipment, mechanical structures and work holding devices.

Scope

This unit/task covers the following:

- Working safely
- Preparing for power plant mechanical, hydraulic and pneumatic maintenance operations
- Ensure upkeep of tools
- Performing power plant mechanical, hydraulic and pneumatic maintenance operations

Elements and Performance Criteria

Working Safely

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety, environmental and other relevant regulations and guidelines at work
- PC2.** adhere to procedures and guidelines for personal protective equipment (ppe) and other relevant safety regulations while performing fabrication and fitting operations
- PC3.** work following laid down procedures and instructions
- PC4.** ensure work area is clean and safe from hazards
- PC5.** ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition
- PC6.** follow all relevant setting up and operating specifications for the products or mechanical equipment being commissioned
- PC7.** follow the defined procedures and set up the equipment correctly ensuring that all operating parameters are achieved

Prepare for power plant mechanical, hydraulic and pneumatic maintenance operations

To be competent, the user/individual on the job must be able to:

- PC8.** obtain job specifications and requirements from valid sources and find out the fault Valid sources: instructions from supervisor, instructions from user of the equipment, condition of end product, person or operator who reported the fault, sensory input (sight, sound, smell, touch), monitoring equipment or gauges
- PC9.** 'seek help of the supervisor or engineer to obtain relevant information for interpretation of drawings, specifications, manufacturers' manuals and other documents needed in the maintenance process'



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- PC10.** need to confirm lock-out/tag-out and zero energy procedures in machines process systems and components as per organizational guidelines before attending any job
- PC11.** follow the work procedure to attend the fault and the tools to be used Tools: e.g. allen key, spanner, torque wrench, pliers, bearing puller, circlipplier, scraper (flat & triangular), etc.
- PC12.** evaluate sensory information to assess and accurately ascertain likely abnormalities Aabnormalities: e.g. change in sound/ vibration/ temperature/ smell / visual impact etc.
- PC13.** perform hands-on inspections like checking tolerances and clearances of machinery, equipment
- PC14.** check for worn, defective, broken or otherwise unacceptable components
- PC15.** check condition, level and temperature of fluids according to manufacturers recommendations
- PC16.** identify common types of metals by examining chips, spark test and magnet test
- PC17.** apply monitoring or testing procedures to help in the fault diagnosis using a range of test equipment Monitoring or testing procedures: alignment checks, force/pressure checks (e.g. spring pressure, hydraulic or pneumatic pressures), leakage, vibration, thermal checks (e.g. bearings, friction surfaces), movement checks (e.g. travel, clearance, levers, links), visual checks Test equipment: measuring instruments/devices, thermal indicators, dial test indicators, audio test devices, torque measuring devices, self-diagnostic equipment, other specific test equipment
- PC18.** relate previous reports/records of similar fault conditions to identify patterns and history
- PC19.** erect and use scaffolding upto 6 meters height using cup lock scaffolding materials as per standard procedures
- PC20.** ensures tool are fit for use and used operated in a safe manner, efficiently Tools: hand tools (files, hacksaws, chisels and hammers); hand held power tools (portable electric drill, portable grinders, hydraulic wrenches, hydraulic jack etc., measuring tools (micrometers, vernier calipers, feeler gauges, telescoping gauges, protractors, dial indicators, straightedges, height gauges, solid square, combination square, combination set, optical equipment and measuring tape; rigging equipment (block and tackles, chains, wire ropes, nylon slings, eye-bolts, hoist rings, hooks, softeners, snatch blocks, spreader bars, lifting beams and shackles); hoisting/lifting and moving equipment (overhead crane, hoist and jacks); machines (drill, presses, stationary grinders, chop saws etc.)
- PC21.** carry out maintenance activities on various power plant equipment in line with job requirement, and as per organizational standards and manufacturers guidelines. Equipment: prime movers (turbines and internal combustion engines); power transmission assemblies (belt drives, gear drives, chain drives, clutches, brakes, couplings); various power plant equipment (fans and blowers, compressors, pumps, conveying systems, etc.); basic hydraulic and pneumatic system
- PC22.** carry out the maintenance activities in the specified sequence and in an agreed timescale

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- PC23.** perform dismantling processes mechanical equipment using appropriate method or technique in order to replace defective components. Dismantling processes: e.g. release of pressures/force, proof marking of components, removal of components by extraction or pressing, etc. range of components: shafts; couplings; gears; clutches; valves and seats; pistons; splined components; brakes; bearing and seals; fitting keys; springs; diaphragms; cams and followers; chains & sprockets; pulleys and belts; levers and links; slides; rollers; tooling; fluid storage units; fabricated components; wire ropes/cables; housings; actuating mechanisms; structural/operational components; locking & retaining devices (e.g. circlips, pins, lock nuts); covers and casings; integrated modules; other specific components. Methods and techniques: release of pressures/forces, proof marking, extraction, pressing, alignment
- PC24.** inspect components to check that the dismantled components are fit for reuse or due for replacement and identify the need to replace lifed items. Components: e.g. seals, gaskets, o-rings etc.
- PC25.** re-assemble the components using appropriate methods, and adjust them to meet the operating specification. Adjustments: setting working clearance, setting travel, setting backlash in gears, preloading bearings, bearing pressing, lubrication oil/grease to be added
- PC26.** carry out servicing and maintenance techniques as applicable Maintenance techniques: installing, dismantling and reinstalling equipment to unit/sub-assembly level; installing, dismantling and reinstalling units to component level; proof marking/labelling of components; checking components for serviceability; replacing all lifed items (e.g. seals, bearings, gaskets); replacing damaged/defective components; setting, aligning and adjusting replaced components; tightening fastenings to the required torque; making off-load checks before starting up; replenishing oils and greases; safety system checks; functionally testing the completed system; check levelling and alignment
- PC27.** replace or refit basic hydraulic and pneumatic components Components: valves; seals; buckets; cylinders; clamping and positioning components; other basic components
- PC28.** identify requirements for machining, electric or electronic repair and to intimate the supervisor/engineer for arrangement to handover to the relevant personal after following due process for rectification
- PC29.** witness a trial run of the equipment at full power/speed/flow to identify any abnormality of the repaired/installed equipment
- PC30.** confirm that the attended component/equipment gives expected process outcomes
- PC31.** carry out maintenance as per maintenance procedures and plans Procedures and plans: e.g. preventive maintenance (routine inspections, and adjustments); corrective maintenance (activities identified from preventative maintenance activities); supervision and guidance to the junior team members; etc.
- PC32.** deal with equipment malfunction and rectify faults during the breakdown servicing process as appropriate. Breakdown categories: intermittent problem, partial failure/out-of specification output, complete breakdowns, preventive maintenance
- PC33.** adhere to a routine schedule of maintenance activities to prevent faults
- PC34.** apply predictive maintenance tests for early detection of equipment defects Tests for predictive maintenance: non-destructive tests (dye penetrant, based on the characteristics of the machinery, equipment or component being tested), static balancing and alignment
- PC35.** deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve
- PC36.** leave the work area in a safe and tidy condition on completion of the maintenance activities



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** relevant health and safety requirements applicable in the work place
- KU3.** layout of a power plant and the various functions
- KU4.** importance of working in clean and safe environment
- KU5.** own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
- KU6.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU7.** relevant people and their responsibilities within the work area
- KU8.** escalation matrix and procedures for reporting work and employment related issues
- KU9.** documentation and related procedures applicable in the context of employment and work
- KU10.** importance and purpose of documentation in context of employment and work
- KU11.** service request procedures, tools, and techniques
- KU12.** organizational procedure(s) to be adopted for the safe disposal of waste of all types of materials
- KU13.** health and safety requirements, and safe working practices and procedures required for the mechanical maintenance activities undertaken Safe working practices and procedures ensuring the correct isolation of the machine or system before starting of any job on the machine or system; fitting and adjusting machine guards; ensuring personal protective equipment (ppe) to be worn for the maintenance activities e.g. correctly fitting overalls and safety glasses; ensuring long hair is tied back or netted; jewellery or other items that can become entangled in the machinery are removed
- KU14.** hazards associated with the mechanical maintenance activities and how they can be minimized. Hazards: handling oils; greases; stored pressure/force; misuse of tools; using damaged or badly maintained tools and equipment; not following laid-down maintenance procedures
- KU15.** hazards and isolating procedures associated with high pressure fluids, zero energy and stored energy
- KU16.** isolation and lock-off procedures or permit-to-work procedure that applies lock-out, tag-out and zero energy procedures
- KU17.** how to extract and use information from engineering drawings and related specifications in relation to work undertaken with the help of supervisor/engineer when needed
- KU18.** how to interpret imperial and metric systems of measurement, workpiece reference points and system of tolerancing
- KU19.** power plant terminology
- KU20.** the methods of positioning, aligning and securing the workpiece



Qualification Pack

- KU21.** assembly methods, techniques and procedures to be used Methods: assembling components having interference fits (e.g. by pressure, expansion or contraction); securing components using threaded fasteners (e.g. nuts, bolts, machine screws, cap screws); securing components using spring clips (e.g. external circlips, internal circlips, special clips); using locking and retaining devices (e.g. tab washers, locking nuts, wire locks); securing components using rivets (e.g. countersunk, roundhead, blind); applying sealing compounds or adhesives; setting and adjusting components to give correct working parameters (e.g. shimming and packing); torque setting of nuts and bolts; by welding
- KU22.** how the components are to be aligned, adjusted and positioned prior to securing them, and the tools and equipment Tools and equipment: clamping direct to machine table, pneumatic or magnetic table; machine vice (e.g. plain, swivel, universal); angle plate; vee block and clamps; fixtures; chucks (e.g. 3, 4 jaw); magnetic chucks; in a bench vice; collets
- KU23.** various mechanical fastening devices that are used Fastening devices: nuts; bolts; machine screws; cap screws; clips; pins; locking and retaining devices; rivets
- KU24.** procedure(s) to be followed for investigating the faults, and how to deal with intermittent faults
- KU25.** how to analyze and evaluate possible characteristics and causes of specific faults/problems
- KU26.** types of prime movers such as electric motors, turbines, and internal combustion engines
- KU27.** types of turbines such as gas, wind, water and steam and associated equipment such as boilers
- KU28.** auxiliary systems such as lubrication and cooling
- KU29.** common faults such as loss of rpm and the failure of bearings, seals and lubrication systems
- KU30.** components such as bearings, shafts, seals, couplings, clutches, brakes, chains, belts and gears
- KU31.** types of bearing fits such as press, slide and interference
- KU32.** types of bearing housings such as pillow blocks, split, flange
- KU33.** types of bearings such as radial and axial
- KU34.** anti-friction (rolling element-type) bearing and journal bearing installation and removal
- KU35.** components and tools such as taper locks and pullers
- KU36.** bearing faults such as loss of clearance, overheating, excessive lubrication and lack of lubrication
- KU37.** type of spindles and shafts Types: Universal spindle, Plain shaft, Hollow shaft, crank shaft, cam shaft
- KU38.** seal faults such as leaking, deterioration and improper installation
- KU39.** seal materials compatibility with medium
- KU40.** types of seals such as static, dynamic, mechanical, contact and non-contact
- KU41.** shaft faults such as bent and worn shafts
- KU42.** shaft restoration applications such as shaft straightening, spray welding, peening, knurling and using sleeves
- KU43.** clutch and brake faults such as wear, overheating, excessive vibration and slippage
- KU44.** types and components of couplings, clutches, brakes and fasteners and retainers
- KU45.** coupling faults such as compromised transfer of movement, excessive vibration and worn components



Qualification Pack

- KU46.** indications of component failure such as clutch and brake slippage, excessive heat and vibration
- KU47.** required clearances and tolerances for couplings, clutches and brakes
- KU48.** speed, length, sizing and ratio calculations for belt drive and chain
- KU49.** belt drive and chain drive types and components
- KU50.** manufacturers' specifications such as tension, rpm, capacity limitations and operating conditions
- KU51.** gear faults such as overheating, vibration and excessive noise
- KU52.** gear system components and their installation requirements such as fits, thrust, clearances and tolerances
- KU53.** gear terminology such as pitch diameter, diametral pitch, dedendum, addendum and working depth
- KU54.** installation methods for gear drives such as pressed, keyed, sliding and pinned
- KU55.** lubrication methods such as splash, forced and oil rings
- KU56.** procedure for obtaining replacement parts, materials and other consumables necessary for the maintenance activities
- KU57.** sequence to be adopted for the dismantling/re-assembly of various types of assemblies
- KU58.** methods and techniques used to dismantle/assemble mechanical equipment
Methods and techniques: release of pressures/forces, proof marking, extraction, pressing, alignment
Methods to produce mechanical assemblies: assembling components having interference fits (e.g. by pressure, expansion or contraction); securing components using threaded fasteners (e.g. nuts, bolts, machine screws, cap screws); securing components using spring clips (e.g. external circlips, internal circlips, special clips); using locking and retaining devices (e.g. tab washers, locking nuts, wire locks, special purpose types); securing components using rivets (e.g. countersunk, roundhead, blind, special purpose types); applying sealing compounds or adhesives; electrical bonding of components; setting and adjusting components to give correct working parameters (e.g. shimming and packing); torque setting of nuts and bolts; sby welding
- KU59.** methods of checking components are fit for purpose, and how to identify defects and wear characteristics
- KU60.** monitoring equipment such as temperature probes and thermographic equipment, oil analysis, vibration analysis, ultrasound devices, fluid analysis, infrared thermography and motor current analysis
- KU61.** basic principles of how the equipment functions, operation sequence, the working purpose of individual units/components and how they interact
- KU62.** identification, application, fitting and removal of different types of bearings and gears
- KU63.** how to correctly adjust tension belts and chains
- KU64.** identification and application of different types of locking devices
- KU65.** methods of checking that removed components are fit for purpose, and the need to replace 'lifer' items
- KU66.** uses of measuring equipment
Measuring equipment: external micrometers, vernier/digital/dial caliper, surface finish equipment (e.g. comparison plates, machines), rules, squares, protractors, depth micrometers, depth verniers, feeler gauges, bore/hole gauges, slip gauges, radius/profile gauges, thread gauges, tachometers, torque wrenches, spirit levels



Qualification Pack

- KU67.** how to make adjustments to components/assemblies to ensure they function correctly
Adjustments: setting working clearance, setting travel, setting backlash in gears, preloading bearings, bearing pressing
- KU68.** importance of making 'off-load' checks before running the equipment under power
- KU69.** how to check tools and equipment are free from damage or defects, are in a safe and usable condition, and are configured correctly for the intended purpose
- KU70.** importance of maintenance documentation and/or reports following the maintenance activity
- KU71.** equipment operating and control procedures to be applied during the maintenance activity
Operating and control procedures: organisational guidelines and procedures; equipment manufacturers operating specification/range; recognised compliance agency/body standards or directives; health, safety and environmental requirements; customer standards and requirements
- KU72.** how to use lifting and handling equipment in the maintenance activity
- KU73.** problems associated with the maintenance activity, and how they can be overcome
- KU74.** extent of their own authority and to whom they should report if they have a problem that they cannot resolve
- KU75.** techniques of checking dimensional accuracy
- KU76.** how to check the workpiece and the measuring equipment that is used
- KU77.** need to check that the measuring equipment is within current calibration dates, and that the instruments are correctly zeroed
- KU78.** basic hydraulic systems and components
- KU79.** auxiliary components such as coolers, heaters and accumulators
- KU80.** basic rigging practice
- KU81.** viscosity of fluids
- KU82.** installation procedures for hydraulic systems and components
- KU83.** hydraulic system faults such as loss of pressure, cavitations, contamination of fluid, aeration, leaks, loss of movement and speed, and overheating
- KU84.** installation procedures for basic types of pneumatic and vacuum systems and components
- KU85.** pneumatic system faults such as leaks, loss of movement and speed, and overheating
- KU86.** benefits of RCM methodology
- KU87.** preventive and predictive maintenance programs and schedules
- KU88.** NDT techniques such as dye penetrant, magnetic particle, radiography and ultrasonic
- KU89.** types of imbalance such as static, dynamic and coupled
- KU90.** basic metallurgy
- KU91.** properties and characteristics of common types of metals and materials use in the trade
- KU92.** ways in which a metal can fail
- KU93.** compatibility of metals and of other materials
- KU94.** heat treatment procedures such as annealing, hardening, tempering and normalizing of metals
- KU95.** when to act on their own initiative and when to seek help and advice from others
- KU96.** importance of leaving the work area and equipment in a safe and clean condition on completion of the machining and fitting activities



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in local language
- GS2.** check and clarify task-related information
- GS3.** liaise with appropriate authorities using correct protocol
- GS4.** convey and share technical information clearly using appropriate language
- GS5.** communicate with people in respectful form and manner in line with organizational protocol
- GS6.** undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
- GS7.** identify and draw various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Compound shapes: involving squares, rectangles, triangles, circles, semicircles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder
- GS8.** use appropriate measuring techniques and units of measurement
- GS9.** use appropriate units and number systems to express degree of accuracy Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
- GS10.** calculations related to force and pressure relevant to operating/testing the machines to be maintained
- GS11.** participate in on-the-job and other learning, training and development interventions and assessments
- GS12.** clarify task related information with appropriate personnel or technical adviser
- GS13.** seek to improve and modify own work practices
- GS14.** maintain current knowledge of application standards, legislation, codes of practice and product/process developments
- GS15.** identify problems with work planning, procedures, output and behavior and their implications
- GS16.** prioritize and plan for problem solving
- GS17.** communicate problems appropriately to others
- GS18.** identify sources of information and support for problem solving
- GS19.** seek assistance and support from other sources to solve problems
- GS20.** identify effective resolution techniques
- GS21.** select and apply resolution techniques
- GS22.** seek evidence for problem resolution
- GS23.** plan, prioritize and sequence work operations as per job requirements
- GS24.** organize and analyze information relevant to work
- GS25.** basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
- GS26.** undertake and express new ideas and initiatives to others



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- GS27.** modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
- GS28.** ones competencies in new and different situations and contexts to achieve more
- GS29.** ones competencies in new and different situations and contexts to achieve more
- GS30.** avoid and manage distractions to be disciplined at work
- GS31.** Manage own time for achieving better results
- GS32.** work in a team in order to achieve better results
- GS33.** identify and clarify work roles within a team
- GS34.** communicate and cooperate with others in the team for better results
- GS35.** seek assistance from fellow team members

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working Safely</i>	5	16	-	-
PC1. comply with health and safety, environmental and other relevant regulations and guidelines at work	1	3	-	-
PC2. adhere to procedures and guidelines for personal protective equipment (ppe) and other relevant safety regulations while performing fabrication and fitting operations	1	3	-	-
PC3. work following laid down procedures and instructions	1	2	-	-
PC4. ensure work area is clean and safe from hazards	-	2	-	-
PC5. ensure that all tools, equipment, power tool cables, extension leads are in a safe and usable condition	-	2	-	-
PC6. follow all relevant setting up and operating specifications for the products or mechanical equipment being commissioned	1	2	-	-
PC7. follow the defined procedures and set up the equipment correctly ensuring that all operating parameters are achieved	1	2	-	-
<i>Prepare for power plant mechanical, hydraulic and pneumatic maintenance operations</i>	20	59	-	-
PC8. obtain job specifications and requirements from valid sources and find out the fault Valid sources: instructions from supervisor, instructions from user of the equipment, condition of end product, person or operator who reported the fault, sensory input (sight, sound, smell, touch), monitoring equipment or gauges	-	1	-	-
PC9. 'seek help of the supervisor or engineer to obtain relevant information for interpretation of drawings, specifications, manufacturers' manuals and other documents needed in the maintenance process'	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. need to confirm lock-out/tag-out and zero energy procedures in machines process systems and components as per organizational guidelines before attending any job	1	2	-	-
PC11. follow the work procedure to attend the fault and the tools to be used Tools: e.g. allen key, spanner, torque wrench, pliers, bearing puller, circlipplier, scraper (flat & triangular), etc.	1	2	-	-
PC12. evaluate sensory information to assess and accurately ascertain likely abnormalities Abnormalities: e.g. change in sound/ vibration/ temperature/ smell / visual impact etc.	1	2	-	-
PC13. perform hands-on inspections like checking tolerances and clearances of machinery, equipment	1	3	-	-
PC14. check for worn, defective, broken or otherwise unacceptable components	1	2	-	-
PC15. check condition, level and temperature of fluids according to manufacturers recommendations	1	2	-	-
PC16. identify common types of metals by examining chips, spark test and magnet test	1	1	-	-
PC17. apply monitoring or testing procedures to help in the fault diagnosis using a range of test equipment Monitoring or testing procedures: alignment checks, force/pressure checks (e.g. spring pressure, hydraulic or pneumatic pressures), leakage, vibration, thermal checks (e.g. bearings, friction surfaces), movement checks (e.g. travel, clearance, levers, links), visual checks Test equipment: measuring instruments/devices, thermal indicators, dial test indicators, audio test devices, torque measuring devices, self-diagnostic equipment, other specific test equipment	1	2	-	-
PC18. relate previous reports/records of similar fault conditions to identify patterns and history	1	2	-	-
PC19. erect and use scaffolding upto 6 meters height using cup lock scaffolding materials as per standard procedures	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. ensures tool are fit for use and used operated in a safe manner, efficiently Tools: hand tools (files, hacksaws, chisels and hammers); hand held power tools (portable electric drill, portable grinders, hydraulic wrenches, hydraulic jack etc., measuring tools (micrometers, vernier calipers, feeler gauges, telescoping gauges, protractors, dial indicators, straightedges, height gauges, solid square, combination square, combination set, optical equipment and measuring tape; rigging equipment (block and tackles, chains, wire ropes, nylon slings, eye-bolts, hoist rings, hooks, softeners, snatch blocks, spreader bars, lifting beams and shackles); hoisting/lifting and moving equipment (overhead crane, hoist and jacks); machines (drill, presses, stationary grinders, chop saws etc.)	-	2	-	-
PC21. carry out maintenance activities on various power plant equipment in line with job requirement, and as per organizational standards and manufacturers guidelines. Equipment: prime movers (turbines and internal combustion engines); power transmission assemblies(belt drives, gear drives, chain drives, clutches,brakes, couplings); various power plant equipment (fans and blowers,compressors, pumps, conveying systems, etc.); basic hydraulic and pneumatic system	-	2	-	-
PC22. carry out the maintenance activities in the specified sequence and in an agreed timescale	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. perform dismantling processes mechanical equipment using appropriate method or technique in order to replace defective components. Dismantling processes: e.g. release of pressures/force, proof marking of components, removal of components by extraction or pressing, etc. range of components: shafts; couplings; gears; clutches; valves and seats; pistons; splined components; brakes; bearing and seals; fitting keys; springs; diaphragms; cams and followers; chains & sprockets; pulleys and belts; levers and links; slides; rollers; tooling; fluid storage units; fabricated components; wire ropes/cables; housings; actuating mechanisms; structural/operational components; locking & retaining devices (e.g. circlips, pins, lock nuts); covers and casings; integrated modules; other specific components. Methods and techniques: release of pressures/forces, proof marking, extraction, pressing, alignment	1	3	-	-
PC24. inspect components to check that the dismantled components are fit for reuse or due for replacement and identify the need to replace lifed items. Components: e.g. seals, gaskets, o-rings etc.	1	2	-	-
PC25. re-assemble the components using appropriate methods, and adjust them to meet the operating specification. Adjustments: setting working clearance, setting travel, setting backlash in gears, preloading bearings, bearing pressing, lubrication oil/grease to be added	1	3	-	-
PC26. carry out servicing and maintenance techniques as applicable Maintenance techniques: installing, dismantling and reinstalling equipment to unit/sub-assembly level; installing, dismantling and reinstalling units to component level; proof marking/labelling of components; checking components for serviceability; replacing all lifed items (e.g. seals, bearings, gaskets); replacing damaged/defective components; setting, aligning and adjusting replaced components; tightening fastenings to the required torque; making off-load checks before starting up; replenishing oils and greases; safety system checks; functionally testing the completed system; check levelling and alignment	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. replace or refit basic hydraulic and pneumatic components Components: valves; seals; buckets; cylinders; clamping and positioning components; other basic components	-	2	-	-
PC28. identify requirements for machining, electric or electronic repair and to intimate the supervisor/engineer for arrangement to handover to the relevant personal after following due process for rectification	1	1	-	-
PC29. witness a trial run of the equipment at full power/speed/flow to identify any abnormality of the repaired/installed equipment	-	2	-	-
PC30. confirm that the attended component/equipment gives expected process outcomes	-	2	-	-
PC31. carry out maintenance as per maintenance procedures and plans Procedures and plans: e.g. preventive maintenance (routine inspections, and adjustments); corrective maintenance (activities identified from preventative maintenance activities); supervision and guidance to the junior team members; etc.	1	3	-	-
PC32. deal with equipment malfunction and rectify faults during the breakdown servicing process as appropriate. Breakdown categories: intermittent problem, partial failure/out-of specification output, complete breakdowns, preventive maintenance	1	2	-	-
PC33. adhere to a routine schedule of maintenance activities to prevent faults	1	1	-	-
PC34. apply predictive maintenance tests for early detection of equipment defects Tests for predictive maintenance: non-destructive tests (dye penetrant, based on the characteristics of the machinery, equipment or component being tested), static balancing and alignment	1	2	-	-
PC35. deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve	-	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC36. leave the work area in a safe and tidy condition on completion of the maintenance activities	-	2	-	-
NOS Total	25	75	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N0302
NOS Name	Perform maintenance activities on power plant equipment and machinery
Sector	Power
Sub-Sector	Generation
Occupation	Thermal Power Plant Equipment Erection & Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	13/09/2017
Next Review Date	31/03/2022
NSQC Clearance Date	NA



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** give information to others clearly, at a pace and in a manner that helps them to understand
- PC4.** display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** display appropriate communication etiquette while working
- PC7.** display active listening skills while interacting with others at work
- PC8.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** demonstrate responsible and disciplined behaviors at the workplace
- PC10.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and co-ordinate with in the organization
- KU6.** importance of effective communication in the workplace
- KU7.** importance of teamwork in organizational and individual success
- KU8.** various components of effective communication
- KU9.** key elements of active listening
- KU10.** value and importance of active listening and assertive communication



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- KU11.** barriers to effective communication
- KU12.** importance of tone and pitch in effective communication
- KU13.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** importance of ethics for professional success
- KU16.** importance of discipline for professional success
- KU17.** what constitutes disciplined behavior for a working professional
- KU18.** common reasons for interpersonal conflict
- KU19.** importance of developing effective working relationships for professional success
- KU20.** expressing and addressing grievances appropriately and effectively
- KU21.** importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge
- GS2.** note down observations (if any) related to the operation/maintenance
- GS3.** read and interpret the process required for different types of manuals
- GS4.** read and interpret the flowchart of all parts of an assembly
- GS5.** read manuals and documents to understand the product-details & how they can be used
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor
- GS7.** effectively communicate with the team members
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor
- GS9.** communicate clearly with the colleague on the issues faced during query/fault
- GS10.** follow colleague/contractor rule-based decision making process
- GS11.** take decisions with systematic course of actions and/or response
- GS12.** planning and organization of tasks to meet deadlines
- GS13.** build customer relationships and use customer centric approach
- GS14.** seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly
- GS15.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing
- GS16.** critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>working with others</i>	30	70	-	-
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023



Qualification Pack

PSS/N2001: Use basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** identify job-site hazardous work and state possible causes of risk or accident in the workplace
- PC5.** follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),
- PC6.** follow warning signs (danger, out of service, etc.) while working with electrical systems
- PC7.** use standard safe working practices when working at heights, confined areas and trenches
- PC8.** test any electrical equipment and system using insulated testing devices before touching them
- PC9.** ensure positive isolation of electrical equipment & system as per given standards
- PC10.** recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed
- PC11.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC12.** state methods of accident prevention in the work environment of the job role
- PC13.** state location of general health and safety equipment in the workplace
- PC14.** inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder
- PC15.** lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa
- PC16.** inspect grid station and its equipment routinely for any signs of oil and water leakage
- PC17.** store flammable materials and machine lubricating oil safely and correctly
- PC18.** check that the emission and pollution control devices are working properly in line with environmental policy standards
- PC19.** ensure proper working condition of battery and battery charger
- PC20.** maintain electrolyte level of each cell with distilled water
- PC21.** maintain proper ventilation in battery room
- PC22.** apply good housekeeping practices at all times
- PC23.** identify common hazard signs displayed in various areas



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PC24. retrieve and/or point out documents that refer to health and safety in the workplace

PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly

Fire Safety

To be competent, the user/individual on the job must be able to:

PC26. use the various appropriate fire extinguishers on different types of fires correctly

PC27. distinguish between various types of fire

PC28. demonstrate rescue techniques applied during fire hazard

PC29. demonstrate good housekeeping in order to prevent fire hazards

PC30. demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC31. demonstrate how to free a person from electrocution

PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC33. demonstrate basic techniques of bandaging

PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC37. demonstrate the artificial respiration and the cpr process

PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work

PC39. write accident/incident report or dictate a report to another person, and send report to person responsible

PC40. demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. escalation matrix and procedures for reporting work and employment related issues

KU5. meaning of hazards and risks

KU6. health and safety hazards commonly present in the work environment and related precautions

KU7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible



Qualification Pack

- KU8.** possible causes of risk and accident
- KU9.** methods of accident prevention
- KU10.** safe working practices when working with tools and machines
- KU11.** safe working practices while working at various hazardous sites
- KU12.** where to find all the general health and safety equipment in the workplace
- KU13.** various dangers associated with the use of electrical equipment
- KU14.** positive isolation of electrical equipment and system
- KU15.** safe handling and disposal of hazardous power plant wastes
- KU16.** use of emission and pollution control devices and measures taken to control pollution
- KU17.** various safety procedures and equipment used to work at heights, trenches and confined places
- KU18.** safe working practices specific to working with electrical equipment & system e.g. lock out/tag out, ptw, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU20.** importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
- KU21.** precautionary activities taken to prevent fire accident
- KU22.** various causes of fire
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire
- KU26.** emergency rescue techniques applied during a fire hazard
- KU27.** various types of safety signs and what they mean
- KU28.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the information communicated by the officer incharge.
- GS2.** note down observations (if any) related to the operation/maintenance.
- GS3.** read and interpret the process required for different types of manuals for maintenance.
- GS4.** read and interpret the flowchart of all parts of an assembly.
- GS5.** read manuals and documents to understand the product-details & how they can be used.
- GS6.** discuss task lists, schedules and activities with the colleague/supervisor.
- GS7.** effectively communicate with the team members.
- GS8.** attentively listen and comprehend the information given by the colleague/supervisor/contractor.
- GS9.** communicate clearly with the colleague on the issues faced during query/fault.
- GS10.** follow colleague/contractor rule-based decision making process.
- GS11.** take decisions with systematic course of actions and/or response.



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- GS12.** planning and organization of tasks to meet deadlines.
- GS13.** build customer relationships and use customer centric approach.
- GS14.** seek and comprehend operation related inputs for clarification
- GS15.** find ways of modifying difficult operating stages to make it operation friendly
- GS16.** work systematically and logically to resolve the issues and identify causation and anticipate unexpected results.
- GS17.** quick approach and solution towards faults repairing.
- GS18.** critically evaluate operation parameters in relation to system normality
- GS19.** develop a holistic and comprehensive profile of grid station on segregated discrete process.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	11	46	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions	-	3	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	-	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	-	2	-	-
PC4. identify job-site hazardous work and state possible causes of risk or accident in the workplace	1	1	-	-
PC5. follow electrical safe working procedures such as tag out/lock out and display ptw (permit to work),	1	2	-	-
PC6. follow warning signs (danger, out of service, etc.) while working with electrical systems	1	1	-	-
PC7. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC8. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC9. ensure positive isolation of electrical equipment & system as per given standards	1	2	-	-
PC10. recognize any abnormalities in electrical equipment or system installed alarm annunciation and/or noticing parameters from gauge/ indicator installed	1	2	-	-
PC11. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	1	-	-
PC12. state methods of accident prevention in the work environment of the job role	-	2	-	-
PC13. state location of general health and safety equipment in the workplace	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inspect for faults, set up and safely use of scaffolds and elevated platforms and ladder	-	2	-	-
PC15. lift, carry and transport heavy objects & tools safely using correct procedures from storage to workplace and vice versa	-	2	-	-
PC16. inspect grid station and its equipment routinely for any signs of oil and water leakage	-	2	-	-
PC17. store flammable materials and machine lubricating oil safely and correctly	-	2	-	-
PC18. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	1	-	-
PC19. ensure proper working condition of battery and battery charger	1	1	-	-
PC20. maintain electrolyte level of each cell with distilled water	-	2	-	-
PC21. maintain proper ventilation in battery room	-	1	-	-
PC22. apply good housekeeping practices at all times	1	2	-	-
PC23. identify common hazard signs displayed in various areas	-	2	-	-
PC24. retrieve and/or point out documents that refer to health and safety in the workplace	-	2	-	-
PC25. inform relevant authorities about any abnormal situation/behavior of any equipment/system promptly	-	3	-	-
Fire Safety	4	9	-	-
PC26. use the various appropriate fire extinguishers on different types of fires correctly	1	1	-	-
PC27. distinguish between various types of fire	1	2	-	-
PC28. demonstrate rescue techniques applied during fire hazard	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate good housekeeping in order to prevent fire hazards	-	2	-	-
PC30. demonstrate the correct use of a fire extinguisher	1	2	-	-
<i>Emergencies, rescue and first-aid procedures</i>	9	21	-	-
PC31. demonstrate how to free a person from electrocution	1	2	-	-
PC32. administer appropriate first aid to victims where required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	-	3	-	-
PC33. demonstrate basic techniques of bandaging	1	2	-	-
PC34. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC35. perform and organize loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-
PC36. administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	2	-	-
PC37. demonstrate the artificial respiration and the cpr process	1	2	-	-
PC38. participate in emergency procedures emergency procedures: raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC39. write accident/incident report or dictate a report to another person, and send report to person responsible	1	2	-	-
PC40. demonstrate correct method to move injured people and others during an emergency	1	2	-	-
NOS Total	24	76	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2001
NOS Name	Use basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	14/04/2024
NSQC Clearance Date	31/03/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N0301.Install power plant mechanical equipment at site	24	76	-	-	100	40
PSS/N0302.Perform maintenance activities on power plant equipment and machinery	25	75	-	-	100	40
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2001.Use basic health and safety practices for power related work	24	76	-	-	100	15
Total	103	297	-	-	400	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
OEE	Overall Equipment Effectiveness



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.